

Problem 1.13

Discounting is the process of finding the present value of an amount of cash at some future date. By the present value we mean the principal that must be invested now in order to achieve a desired accumulated value over a specified period of time. Find the present value of \$100 in five years time if the annual compound interest is 12%.

Solution

Using the compound interest formula

$$A = P(1 + i)^t,$$

we solve for the present value P :

$$P = \frac{A}{(1 + i)^t}.$$

Substitute the given values $A = 100$, $i = 0.12$, and $t = 5$:

$$P = \frac{100}{(1.12)^5}.$$

Calculating,

$$P \approx \frac{100}{1.76234168} \approx 56.74.$$

Hence, the present value is approximately

$$\boxed{\$56.74}.$$